

Virtualized Hardware-Style Next Generation Firewall With AI-Assisted Intrusion Detection and DPI

PROJECT REPORT

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of

Bachelor of Technology

in

Computer Science and Engineering (Cyber Security)



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Security)**

College Of Engineering Kallooppara

Kerala

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Declaration

We undersigned hereby declare that the project report **Virtualized Hardware-Style Next Generation Firewall with AI-Assisted Intrusion detection and DPI** submitted for fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Assistant Professor Raj Kumar T.** This submission represents our ideas in our own words and ideas or words of others have been included where we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the university and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other university.

Place: Kallloopara

Date: 13 March 2026

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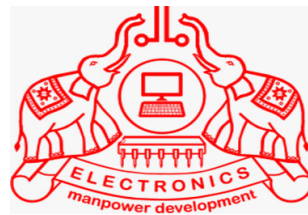
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CERTIFICATE



This is to certify that the project design report entitled **“Virtualized Hardware-Style Next Generation Firewall with AI-Assisted Intrusion detection and DPI”** submitted by **Achala A S (PTA22CC005)**, **Aswin Manoj (PTA22CC024)**, **Hisham Faizal EK (PTA22CC038)**, **Sabari S (PTA22CC054)**, in fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Cyber Security) of APJ Abdul Kalam Technological University is a bonafide work carried out by them under our guidance and supervision. The report in any form has not been submitted to any other University or Institute for any purpose.

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Abstract

With the exponential growth of cyber threats, traditional firewalls that rely solely on IP- and port-based filtering are increasingly inadequate for protecting modern, dynamic network environments. This project proposes the development of a virtualized Next Generation Firewall (NGFW) appliance that runs as a dedicated security gateway virtual machine (VM) with multiple network interfaces (WAN/LAN), ensuring that all client traffic is routed through it. This approach provides a hardware-like deployment experience using commodity virtualization platforms.

The appliance integrates core firewall features such as IP/port filtering, protocol-aware rules, and stateful packet inspection to enforce baseline network policies. Security depth is enhanced through Deep Packet Inspection (DPI) of application-layer protocols such as HTTP, DNS, and FTP to detect malicious patterns, blocked domains, and suspicious file transfer activities.

An SSL/TLS handling module identifies encrypted HTTPS flows, extracts certificate metadata (issuer, validity period, and Common Name), and optionally supports proxy-based decryption for controlled content inspection in laboratory environments. For proactive defense, an Intrusion Detection and Prevention System (IDS/IPS) layer combines signature- and rule-based detection (e.g., SYN flood recognition) with machine learning models such as Random Forest and Support Vector Machine (SVM) for real-time anomaly detection.

Additionally, a Threat Intelligence Integration service ingests public IP and domain blocklists to automatically enforce dynamic reputation-based controls. Operational visibility and control are provided through an integrated web-based management GUI hosted within the firewall VM. This interface enables administrators to visualize network traffic, manage firewall rules, review alerts, examine SSL certificate details, and generate security reports.

By packaging advanced NGFW capabilities into a self-contained virtual firewall appliance with an embedded management interface, this project delivers a scalable, low-cost, and academically demonstrable platform for modern network security research, education, and controlled deployment.

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List of Abbreviations

ML	Machine Learning
AI	Artificial Intelligence
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
TCP	Transmission Control Protocol
UDP	User Datagram Protocol

Chapter 1

Introduction

1.1 Current State of Firewalls

Firewalls form the backbone of modern network security, serving as a protective barrier between trusted internal networks and untrusted external environments such as the Internet. Traditional firewalls rely on static rule-based filtering, inspecting packets based on IP addresses, ports, and protocols.

Over time, Next-Generation Firewalls (NGFWs) have evolved to incorporate advanced features such as Deep Packet Inspection (DPI), Intrusion Detection and Prevention Systems (IDS/IPS), and application-layer visibility. These capabilities enable firewalls to identify specific applications and enforce more granular security policies.

Modern NGFWs can also perform SSL/TLS inspection to analyze encrypted traffic, integrate sand-boxing and malware protection mechanisms, and connect with global threat intelligence feeds for real-time updates. These advancements allow NGFWs to provide more context-aware and adaptive protection across enterprise, cloud, and hybrid environments while maintaining consistent security policies and compliance standards.

1.2 Challenges

Despite significant technological progress, firewalls face growing challenges in today's dynamic cyber security landscape. Attackers increasingly use sophisticated techniques such as zero-day exploits, polymorphic malware, encrypted command-and-control traffic, and AI-driven attack automation to evade detection.

The rapid growth of cloud computing, virtualization, IoT devices, and remote work has further expanded the attack surface, creating complex and distributed network infrastructures that traditional firewalls struggle to secure.

Additionally, more than 90 percent of network traffic is now encrypted, making it extremely difficult to inspect and analyze without introducing performance bottlenecks

or privacy concerns. Cyber criminals exploit this by hiding malicious payloads within encrypted tunnels, bypassing conventional DPI mechanisms.

Static rule-based systems cannot adapt quickly to evolving threats, while attackers employ advanced evasion techniques such as packet fragmentation, protocol tunneling, and traffic obfuscation to remain undetected. Furthermore, maintaining low latency while processing massive volumes of data and applying complex inspection rules remains a major technical challenge for real-time protection systems.

1.3 Limitations of Traditional Vulnerability Detection Systems

- Most existing Next-Generation Firewalls rely heavily on signature-based detection mechanisms, which identify threats based on previously known attack patterns or signatures. Although this approach is effective against familiar malware or intrusion types, it fails to detect zero-day attacks, polymorphic viruses, and new threat variants that do not match existing signatures. This limitation leaves networks vulnerable to emerging or rapidly evolving attack vectors.

- Commercial NGFW solutions are often closed-source and vendor-controlled, offering limited transparency regarding how their AI models and detection algorithms operate. Security teams typically have little control over model retraining, feature customization, or integration with open-source security tools. As a result, organizations must entirely depend on vendor updates and threat definitions.

- Many advanced firewall capabilities—such as behavioral threat analysis, honeypot-based deception systems, and ransomware protection—are often available only as additional premium modules. This increases the overall cost of deployment and makes comprehensive protection accessible mainly to large enterprises.

- As network traffic volumes continue to grow, existing firewalls struggle to balance deep inspection with system performance. Real-time analysis of encrypted traffic, application-based filtering, and AI-based threat detection require substantial computational resources. This may result in increased latency, reduced throughput, or hardware bottlenecks in high-speed network environments.

- Although AI integration has improved threat detection capabilities, many NGFWs rely on static or vendor-controlled models that cannot dynamically adapt to new threats. These models often function as black boxes, making it difficult for security teams to understand the reasoning behind specific detection decisions.

- Since more than 90 percent of global Internet traffic is encrypted, many firewalls face difficulties in inspecting SSL/TLS communications effectively. Performing full decryption may introduce latency and privacy concerns, while avoiding encrypted inspection creates a significant blind spot that attackers can exploit.

- Modern cyber security environments require integration with Security Information and Event Management (SIEM) systems, automated response platforms, and cloud orchestration tools. However, many existing firewall systems lack open APIs and seamless interoperability, limiting automation and centralized security management.

Chapter 2

Background and Literature Survey

2.1 Next-Generation Firewalls

The evolution of firewall technology has been driven by the increasing sophistication of cyber threats and the growing demand for intelligent and adaptive defense systems. Traditional firewalls primarily rely on static rule-based filtering, which inspects packets based on predefined IP addresses, ports, and protocols. While this approach provides basic network protection, it is insufficient against modern attacks that exploit encryption, behavioral evasion techniques, and zero-day vulnerabilities.

To address these limitations, Next-Generation Firewalls (NGFWs) have emerged, integrating advanced capabilities such as Deep Packet Inspection (DPI), Intrusion Detection and Prevention Systems (IDS/IPS) and application-layer visibility. These technologies enable firewalls to analyze packet payloads, detect suspicious patterns, and enforce granular security policies based on application behavior rather than simple network parameters.

Recent research has also explored the integration of Artificial Intelligence (AI), machine learning techniques, and virtualized architectures into firewall systems to improve detection accuracy, adaptability, and scalability. This section reviews several key studies that have contributed to the development of AI-powered NGFW systems and analyzes their strengths and limitations.

One important work titled “You Do (Not) Belong Here: Detecting DPI Evasion Attacks with Context Learning” proposes a context-aware Deep Packet Inspection approach to detect evasion techniques designed to bypass traditional DPI systems. Instead of analyzing packets individually, the system examines traffic sequences and contextual patterns, allowing it to identify fragmented or obfuscated payloads more effectively. The main strength of this approach is its ability to detect subtle evasion techniques. However, the study mainly focuses on packet inspection mechanisms and does not integrate broader firewall capabilities such as policy enforcement, AI retraining, or deception-based defense strategies.

Another study, “Advanced Threat Detection in Next-Generation Firewalls Using Hybrid Machine Learning Techniques,” introduces a hybrid intrusion detection model that combines supervised and unsupervised learning methods. By integrating classification and clustering algorithms, the framework improves detection accuracy and reduces false positives when identifying sophisticated attacks and zero-day threats. While the hybrid model demonstrates promising detection performance, the work remains largely theoretical and focuses primarily on algorithm design rather than real-world deployment or system integration within a complete firewall architecture.

The research titled “AI-Driven Dynamic Firewall Optimization Using Reinforcement Learning for Anomaly Detection and Prevention” explores the use of reinforcement learning to dynamically optimize firewall policies based on observed network behavior. In this approach, a reinforcement learning agent continuously learns from traffic patterns and adjusts firewall rules in real time to minimize false positives and improve detection capability. Although the concept provides adaptability and automated policy tuning, the study is mainly simulation-based and does not address practical challenges such as integration with DPI engines, scalability, or deployment within enterprise environments.

Another relevant study, “Passive, Transparent, and Selective TLS Decryption for Network Security Monitoring,” focuses on improving visibility into encrypted traffic, which is a major challenge for modern intrusion detection systems. The authors propose a selective key-sharing mechanism where trusted endpoints share encryption keys with a network monitoring system, allowing controlled decryption and analysis of TLS sessions. This approach preserves privacy and session integrity while enabling effective monitoring of encrypted communications. However, the method introduces complexities in key management and does not provide a complete intrusion prevention framework.

The study “Next-Generation Firewall in the Cloud: Advanced Firewall Solutions” discusses how firewall architectures have evolved to support cloud computing and virtualized infrastructures. It highlights the role of Deep Packet Inspection, IDS/IPS mechanisms, and application-layer analysis in protecting modern cloud workloads. The paper also emphasizes the importance of AI and machine learning for improving detection accuracy and automating security policy management. While the study effectively outlines architectural advancements in cloud-based NGFW systems, it provides limited discussion on explainable AI models or behavioral threat detection mechanisms.

From these studies, several important observations can be drawn. Artificial intelligence and machine learning significantly improve the adaptability and detection capability of firewall systems. Hybrid learning approaches and reinforcement learning can enhance anomaly detection, while selective TLS inspection methods improve visibility into encrypted traffic. However, most existing research focuses on isolated components such as DPI evasion detection, machine learning classification, or encrypted traffic monitoring rather than providing a fully integrated firewall framework.

2.2 Key Improvements of the Proposed System

The proposed system, QUOR (Quarantined Unified Operations for Response), introduces several improvements over traditional and existing Next-Generation Firewall solutions. Unlike many commercial systems that rely on vendor-controlled or black-box AI models, QUOR utilizes explainable and retrainable machine learning algorithms such as Random Forest and Support Vector Machine. This approach improves transparency and allows administrators to understand the reasoning behind detection decisions.

QUOR adopts a modular architecture that integrates multiple advanced security components within a unified framework. These components include Deep Packet Inspection (DPI), Intrusion Detection and Prevention Systems (IDS/IPS), TLS/SNI spoof detection, and honeypot-based deception mechanisms. The modular design enables flexible deployment, easier updates, and better scalability for both research and enterprise environments.

In contrast to traditional firewalls that rely primarily on static signature-based detection, QUOR emphasizes behavior-based threat detection. By analyzing traffic patterns, packet sequences, and contextual flow behavior, the system can detect anomalies such as ransomware activity, zero-day exploits, and encrypted attack channels.

Another improvement lies in encrypted traffic analysis. QUOR incorporates TLS/SNI spoof detection and certificate validation mechanisms to identify suspicious encrypted connections and potential man-in-the-middle attacks without requiring full traffic decryption. This approach preserves privacy while still enabling effective monitoring of encrypted communication.

QUOR also integrates a built-in honeypot and deception layer that attracts attackers into controlled environments. This allows the system to collect valuable threat intelligence and observe attacker behavior, which can then be used to improve detection models and strengthen network defenses.

Additionally, QUOR is designed as a virtualized firewall platform that can run on hypervisors such as VMware, VirtualBox, etc. This virtualization-based deployment reduces hardware dependency and makes the system cost-effective and accessible for academic research and small-scale deployments.

2.3 Proposed System Overview

The proposed system, QUOR, is an AI-powered virtualized Next-Generation Firewall designed to provide intelligent and adaptive network protection. The system combines explainable artificial intelligence, modular architecture, and deception-based defense mechanisms within a single integrated platform.

At the core of the system are machine learning models such as Random Forest and Support Vector Machine, which analyze network traffic features and detect anomalies in real time. These models allow the system to identify previously unseen threats and continuously adapt to evolving attack patterns.

QUOR employs a layered security architecture consisting of Deep Packet Inspection, intrusion detection and prevention modules, TLS/SNI spoof detection mechanisms, and honeypot-based deception components. Each module performs a specific function while collaborating with other components to provide comprehensive network protection.

The system also integrates external threat intelligence feeds from platforms such as AbuseIPDB, AlienVault OTX, and VirusTotal. These feeds enrich the system's detection capabilities by providing real-time information about malicious IP addresses, domains, and emerging threats.

A web-based graphical interface enables administrators to monitor network activity, visualize attack patterns, configure system policies, and generate security reports. Through this unified management interface, administrators can control the firewall environment efficiently while maintaining visibility over all security operations.

2.4 Features of the Proposed Next-Generation Firewall

The proposed QUOR firewall includes several advanced features designed to improve security, scalability, and transparency.

One major feature is the Explainable AI-Based IDS/IPS, which uses machine learning model Random Forest to detect anomalies and malicious activities in network traffic. This model provide interpretable results that help administrators understand why certain packets or flows are flagged as threats.

Another key feature is behavior-based threat detection, which monitors traffic patterns and packet sequences to identify abnormal behavior. This allows the system to detect emerging threats such as ransomware propagation and zero-day exploits that may not match existing signatures.

The system also incorporates Deep Packet Inspection (DPI) to analyze packet payloads across multiple protocols including HTTP, DNS, FTP, and SSH. DPI enables the system to detect malicious commands, hidden payloads, and suspicious data transfers within network traffic.

To address encrypted traffic challenges, QUOR includes TLS/SNI spoof detection and encrypted traffic analysis. By validating certificate authenticity and analyzing Server Name Indication (SNI) fields, the system can identify suspicious encrypted connections without performing full decryption.

QUOR also features an integrated honeypot and deception framework that attracts attackers into controlled environments where their behavior can be monitored and analyzed. The intelligence gathered from these interactions is used to strengthen detection models and improve system defense capabilities.

Additionally, the firewall operates in a virtualized environment, enabling deployment on hypervisors or cloud infrastructures. This virtualization-based design improves scalability and reduces reliance on specialized hardware.

Finally, the system provides a real-time web-based dashboard that allows administrators to monitor network traffic, review alerts, analyze threats, and manage firewall configurations through a centralized interface.

2.5 Integration of Firewall Components

QUOR is designed as an integrated security platform where multiple modules collaborate to detect and respond to network threats efficiently.

The packet capture and Deep Packet Inspection module acts as the entry point for network traffic. Tools Scapy is used capture packets and analyze their headers and payloads to identify suspicious behavior.

The extracted traffic data is then processed by the AI-powered intrusion detection and prevention system, which uses machine learning models to classify network behavior as normal or malicious.

An explainable AI engine interprets detection results and provides reasoning behind each classification decision. This transparency allows administrators to verify model behavior and retrain the system using updated datasets.

The TLS/SNI spoof detection module analyzes encrypted sessions and validates SSL/TLS certificates to detect spoofing or man-in-the-middle attacks.

The honeypot and deception layer operates as an isolated environment that captures attacker behavior and feeds the collected intelligence back into the detection models.

External threat intelligence services further enhance detection accuracy by providing real-time updates about malicious IP addresses and domains.

All components are integrated through a centralized web-based graphical interface, which enables monitoring, configuration, and reporting.

2.6 Firewall Implementation Strategy

The implementation of the QUOR firewall follows a structured approach consisting of multiple stages.

The first stage involves data collection and packet capture, where network traffic is

gathered using Scapy. The collected data is preprocessed to extract relevant features for machine learning analysis.

The second stage focuses on AI model development and training. Machine learning algorithms such as Random Forest are trained using labeled datasets to distinguish between normal and malicious traffic.

Next, the trained models are integrated into the firewall architecture along with modules for DPI, intrusion detection, TLS/SNI spoof detection, and honeypot deception.

The system is then deployed in a virtualized environment using platform VirtualBox to ensure scalability and cost-effectiveness.

A web-based dashboard is developed to provide real-time monitoring and management of the firewall system.

Finally, the system undergoes testing and performance evaluation using simulated cyber attacks to measure detection accuracy, latency, and overall system performance. Continuous optimization ensures that the firewall adapts effectively to evolving cyber threats.

Chapter 3

System Design and Architecture

3.1 System Overview

The proposed system, QUOR (Quarantined Unified Operations for Response), is designed as a virtualized Next Generation Firewall (NGFW) capable of monitoring, analyzing, and controlling network traffic between external and internal networks. The system integrates traditional firewall filtering with advanced techniques such as Deep Packet Inspection (DPI), AI-based anomaly detection, encrypted traffic inspection, and threat intelligence verification.

In the proposed architecture, the firewall operates as an intermediate gateway between the internet and internal client devices. All incoming and outgoing network traffic must pass through the firewall before reaching its destination. This allows the system to inspect packets in real time and identify malicious activities before they reach internal systems.

The architecture is implemented in a modular manner where each component performs a specific function such as packet capture, rule enforcement, anomaly detection, TLS inspection, and system monitoring. The overall architecture of the system is illustrated in Figure 3.1.

3.2 Overall System Architecture

The overall system architecture consists of several interconnected modules that together form the complete firewall system. As shown in Figure 3.1, the internet traffic first passes through the network gateway and is then directed to the NGFW module for inspection.

The architecture includes the following major components:

- Core Firewall Engine
- Packet Capture and Traffic Analysis Module
- AI-Based Traffic Analysis Module
- TLS/SNI Inspection Module
- Threat Intelligence and Malware Detection Module
- Logging and Database System
- Web-Based Monitoring Interface

Each module communicates with the others to ensure that suspicious traffic is detected and appropriate actions are taken.

In the system architecture, internal client systems access the internet through the firewall gateway. The firewall inspects traffic using multiple layers of analysis before allowing or blocking the packets.

3.3 Packet Processing Flow

The packet processing flow of the firewall system is illustrated in Figure 3.2. This flow diagram shows how each network packet is inspected and processed by different modules of the system.

When a packet arrives from either the external network or the internal network, the firewall first receives the packet through the network interface. The packet then passes through multiple verification stages before it is either forwarded or dropped.

The first stage involves applying the core firewall rules, where packets are filtered based on IP address, port number, and connection state. Stateful inspection ensures that packets belong to a valid session before allowing further processing.

If the packet violates any predefined rule, it is immediately blocked and recorded in the system logs.

3.4 Firewall Rule Processing

The firewall rule processing module acts as the first security checkpoint in the system. It applies predefined rules based on parameters such as Source and destination IP addresses, Source and destination port numbers, Network protocol type and Connection state. These rules help enforce basic network security policies and prevent unauthorized connections.

If a packet satisfies the firewall rule conditions, it proceeds to the next stage of inspection. If it fails the rule check, the system logs the event and drops the packet to prevent further processing.

3.5 Threat Intelligence and Malware Verification

After passing the firewall rules, the packet is checked against known malware signatures and threat intelligence databases. This step helps identify traffic associated with known malicious IP addresses, domains, or attack patterns.

Threat intelligence feeds from platforms such as AbuseIPDB, AlienVault OTX, or VirusTotal can be used to identify suspicious sources. If a packet is identified as malicious during this stage, the firewall blocks the packet and records the event in the logging system.

In certain cases, suspicious packets may also be redirected to a honeypot port trap where attacker behavior can be monitored and analyzed.

3.6 Deep Packet Inspection and TLS Analysis

Packets that pass the threat intelligence stage are forwarded to the Deep Packet Inspection (DPI) module. DPI performs detailed inspection of packet payloads to detect hidden malicious data or abnormal traffic patterns.

The system also performs TLS/SNI spoof detection to analyze encrypted HTTPS connections. Instead of decrypting the entire communication, the system inspects TLS handshake information such as Server Name Indication (SNI), Certificate issuer and Certificate validity period. This allows the system to detect suspicious encrypted connections or possible spoofing attempts without violating user privacy.

3.7 AI-Based Traffic Analysis

The architecture also integrates an AI-based traffic analysis module implemented using packet analysis tools such as Scapy combined with machine learning algorithms. This module extracts network traffic features such as Packet size, Flow duration, Protocol behavior, Connection frequency, etc.

These features are analyzed using machine learning models such as Random Forest to detect abnormal traffic patterns that may indicate attacks such as port scanning, ransomware communication, or command-and-control traffic.

AI-based analysis improves the ability of the firewall to detect unknown or zero-day attacks that may not match existing signatures.

3.8 Logging and Database Management

All important system activities are recorded in the logging module. The logging system stores information such as Blocked connections, Suspicious packets, AI detection results and Threat intelligence matches.

These logs are stored in a centralized database which allows administrators to review past events and analyze security incidents. Maintaining detailed logs also helps in performing forensic analysis and improving the firewall detection models.

3.9 Web-Based Monitoring and Management Interface

To simplify system management, the firewall includes a web-based graphical user interface (GUI) for monitoring and control. The monitoring dashboard provides real-time information about Network traffic statistics, Detected threats and alerts, Firewall rule activity and System performance.

Administrators can monitor firewall activity by reviewing system logs and analyzing captured traffic data. The logging module records events such as blocked packets, detected threats, and suspicious connections. These logs can be used to analyze traffic patterns and investigate potential security incidents.

3.10 Virtualized Deployment Model

The firewall system is deployed in a virtualized environment, allowing it to operate as a dedicated security gateway without requiring specialized hardware. The system can be deployed using virtualization platforms such as VirtualBox.

In this deployment model, the firewall virtual machine is placed between the internet gateway and the internal network devices. This ensures that all network traffic is routed through the firewall for inspection. Virtualization also allows easy scalability, testing, and integration with research environments.

3.11 Summary of System Architecture

The system design integrates multiple security mechanisms into a single unified architecture. By combining traditional firewall filtering with AI-based anomaly detection, deep packet inspection, encrypted traffic analysis, and threat intelligence integration, the proposed system provides a comprehensive security solution.

The layered architecture ensures that network packets undergo multiple levels of inspection before being allowed into the internal network. This multi-layered design improves the system's ability to detect both known and unknown cyber threats while maintaining efficient network performance.

3.12 Architecture Diagrams

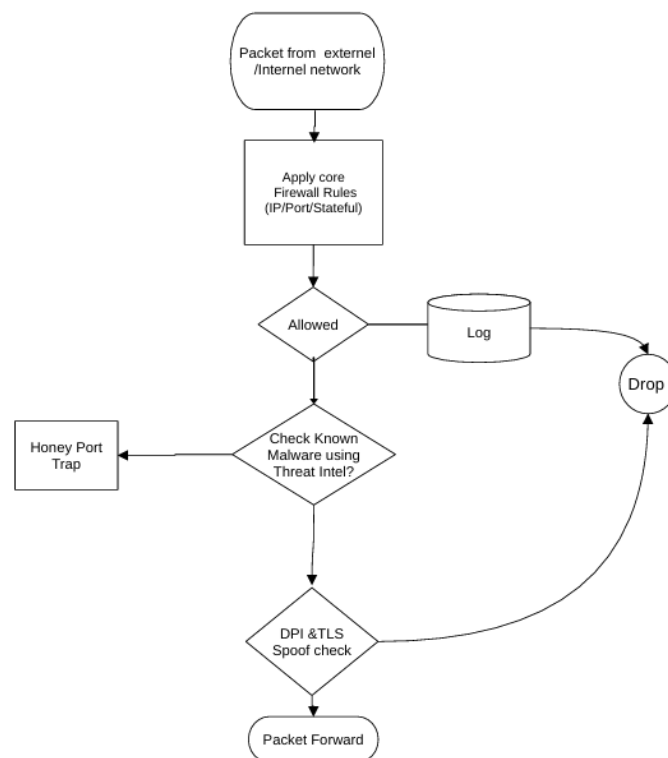


Figure 3.1: Overall System Architecture

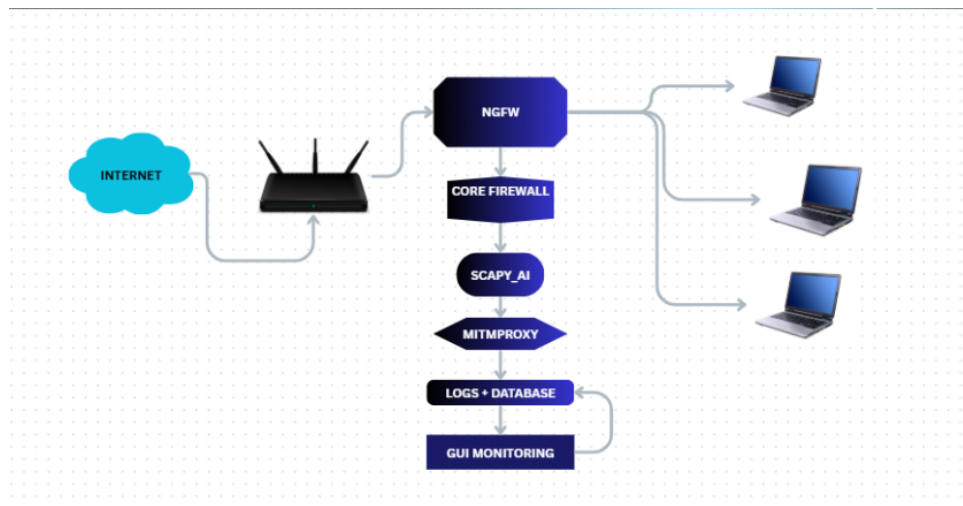


Figure 3.2: Firewall Processing Pipeline

Chapter 4

Implementation Details

4.1 Implementation Overview

The implementation of the QUOR firewall system focuses on combining multiple security mechanisms into a single integrated platform capable of monitoring and analyzing network traffic in real time. The system is developed primarily using Python and several open-source networking tools. The implementation follows a modular structure where each component performs a specific function such as packet capture, firewall rule filtering, traffic inspection, machine learning based detection, encrypted traffic monitoring, and logging. These modules work together sequentially to inspect network packets and determine whether the traffic should be allowed, blocked, or further analyzed.

The overall workflow begins with capturing network packets from the network interface and then applying rule-based filtering. If the packet passes the initial filtering stage, deeper analysis is performed through Deep Packet Inspection, machine learning based anomaly detection, and TLS inspection. Suspicious traffic can also be redirected to a honeypot environment for observation. All detected events are recorded in the logging system for monitoring and analysis.

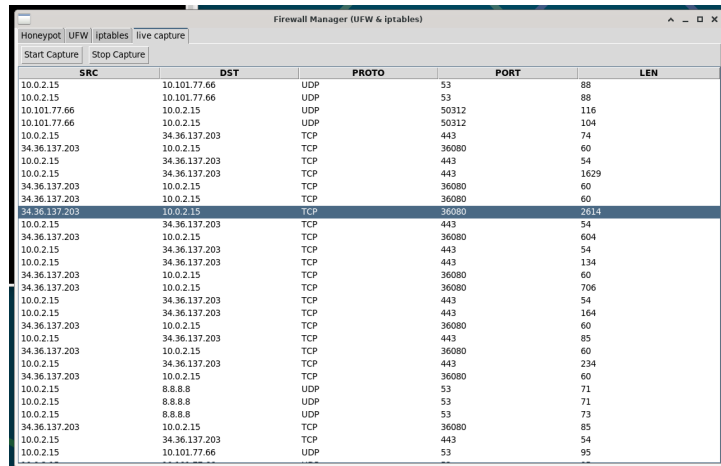
4.2 Packet Capture and Traffic Monitoring

The first step in the implementation process is capturing network packets from the system's network interface. This is achieved using the Scapy library, which provides powerful packet sniffing and manipulation capabilities in Python. Scapy allows the system to observe both incoming and outgoing packets in real time and extract relevant information from them.

When a packet is captured, the system extracts various network attributes such as source IP address, destination IP address, protocol type, port numbers, packet size, and TCP flags. These attributes are useful for both rule-based filtering and machine

learning analysis. By continuously monitoring traffic through the network interface, the firewall gains visibility into all communication occurring between the internal network and external systems.

The captured packet data is then forwarded to the next stage of processing where firewall rules and deeper inspection mechanisms are applied.



SRC	DST	PROTO	PORT	LEN
10.0.2.15	10.101.77.66	UDP	53	88
10.0.2.15	10.101.77.66	UDP	53	88
10.101.77.66	10.0.2.15	UDP	50312	116
10.101.77.66	10.0.2.15	UDP	50312	104
10.0.2.15	34.36.137.203	TCP	443	74
34.36.137.203	10.0.2.15	TCP	36080	60
10.0.2.15	34.36.137.203	TCP	443	54
10.0.2.15	34.36.137.203	TCP	443	1629
34.36.137.203	10.0.2.15	TCP	36080	60
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34.36.137.203	10.0.2.15	TCP	443	2014
10.0.2.15	34.36.137.203	TCP	443	54
34.36.137.203	10.0.2.15	TCP	36080	604
10.0.2.15	34.36.137.203	TCP	443	54
10.0.2.15	34.36.137.203	TCP	443	134
34.36.137.203	10.0.2.15	TCP	36080	60
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10.0.2.15	34.36.137.203	TCP	443	164
34.36.137.203	10.0.2.15	TCP	36080	60
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34.36.137.203	10.0.2.15	TCP	36080	60
10.0.2.15	8.8.8.8	UDP	53	71
10.0.2.15	8.8.8.8	UDP	53	71
10.0.2.15	8.8.8.8	UDP	53	73
34.36.137.203	10.0.2.15	TCP	36080	85
10.0.2.15	34.36.137.203	TCP	443	54
10.0.2.15	10.101.77.66	UDP	53	95

Figure 4.1: Packet Capture and Traffic Monitoring Process

4.3 Firewall Rule Enforcement

Once the packets are captured, the system applies a set of predefined firewall rules to filter traffic based on network policies. This rule enforcement stage acts as the first layer of security in the firewall system. The rules are designed to check parameters such as blocked IP addresses, restricted ports, invalid connection attempts, and abnormal session states.

If a packet violates any of these rules, the system immediately blocks the packet and records the event in the system logs. This prevents unauthorized or suspicious traffic from progressing further into the network. On the other hand, if the packet satisfies the firewall rules, it is forwarded to the next stage for deeper analysis.

Rule-based filtering is important because it reduces unnecessary processing load on the more advanced modules by quickly eliminating clearly malicious or unauthorized traffic.

4.4 Deep Packet Inspection

To detect malicious activity that may not be visible in packet headers alone, the system implements Deep Packet Inspection (DPI). In this stage, the payload of the packet is analyzed in addition to the header information. DPI allows the firewall to observe the actual data being transmitted and detect suspicious communication patterns.

Using Scapy, the system examines traffic associated with protocols such as HTTP, DNS, FTP, and TCP sessions. The inspection process looks for abnormal packet structures, unusual payload sizes, or unexpected protocol behavior that may indicate malicious activity.

During this process, several network traffic features are extracted from the packets. These features are later used by the machine learning module to classify the behavior of the network traffic. Deep packet inspection therefore plays an important role in identifying threats that cannot be detected using simple rule-based filtering.

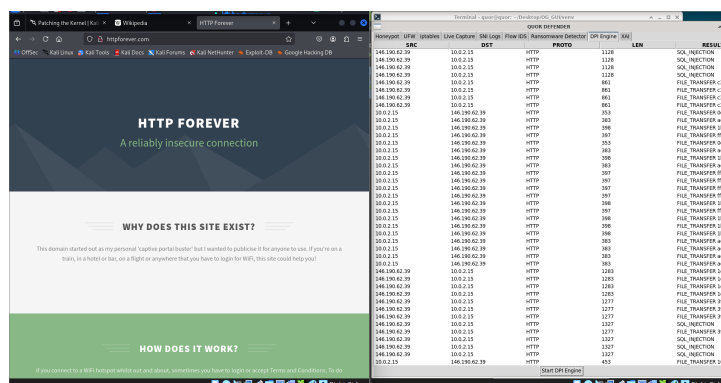


Figure 4.2: Deep Packet Inspection

4.5 Flow-Based Intrusion Detection System

In addition to rule-based filtering and deep packet inspection, the QUOR firewall system incorporates a Flow-Based Intrusion Detection System (Flow IDS) to identify suspicious network behavior using machine learning techniques. Unlike packet-level inspection, Flow IDS analyzes network traffic at the flow level, which represents a sequence of packets exchanged between two endpoints during a communication session.

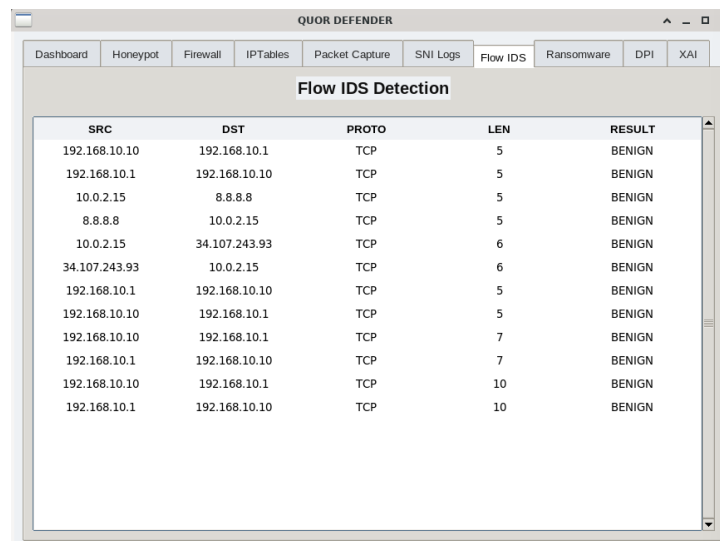
The system groups captured packets into flows based on the standard five-tuple parameters: source IP address, destination IP address, source port, destination port, and protocol type. Each flow represents a logical connection between two devices and provides a summarized view of the network communication.

For every generated flow, the system extracts a set of statistical and behavioral features that describe the characteristics of the traffic. These features include packet count,

total bytes transferred, flow duration, average packet size, and packet inter-arrival times. These values provide useful indicators for distinguishing between normal and abnormal network activity.

Once the features are extracted, they are processed using a data normalization step to ensure compatibility with the trained machine learning model. The normalized feature vector is then passed to the machine learning based detection module, which classifies the flow as either benign or malicious.

The Flow IDS module operates continuously in real time, analyzing newly created flows as network traffic passes through the firewall system. When suspicious or malicious behavior is detected, the system generates an alert and records the event in the logging system for further analysis. By analyzing traffic patterns at the flow level, the Flow IDS enhances the overall security capabilities of the firewall system and helps detect advanced threats that may bypass traditional rule-based filtering mechanisms.



The screenshot shows the 'QUOR DEFENDER' application window with the 'Flow IDS' tab selected. The 'Flow IDS Detection' section displays a table with the following data:

SRC	DST	PROTO	LEN	RESULT
192.168.10.10	192.168.10.1	TCP	5	BENIGN
192.168.10.1	192.168.10.10	TCP	5	BENIGN
10.0.2.15	8.8.8.8	TCP	5	BENIGN
8.8.8.8	10.0.2.15	TCP	5	BENIGN
10.0.2.15	34.107.243.93	TCP	6	BENIGN
34.107.243.93	10.0.2.15	TCP	6	BENIGN
192.168.10.1	192.168.10.10	TCP	5	BENIGN
192.168.10.10	192.168.10.1	TCP	5	BENIGN
192.168.10.10	192.168.10.1	TCP	7	BENIGN
192.168.10.1	192.168.10.10	TCP	7	BENIGN
192.168.10.10	192.168.10.1	TCP	10	BENIGN
192.168.10.1	192.168.10.10	TCP	10	BENIGN

Figure 4.3: Flow-Based Intrusion Detection System

4.6 Machine Learning Based Ransomware Detection

In addition to rule-based detection, the system incorporates a machine learning component to identify suspicious traffic patterns that may indicate ransomware activity or other forms of malicious behavior. A Random Forest classification model is used for this purpose.

The machine learning model is trained using network traffic features that represent both normal and malicious activity. Some of the important features used during training include packet size, connection duration, flow behavior, packet frequency, and TCP flag patterns. These features allow the model to learn differences between legitimate network behavior and potentially harmful traffic.

When the firewall processes a packet during runtime, the extracted features are provided to the trained model. The model then predicts whether the observed traffic is normal or malicious. If the system detects suspicious behavior, the connection may be blocked and the event is recorded in the system logs. This machine learning based approach allows the firewall to identify threats that may not match previously known signatures.

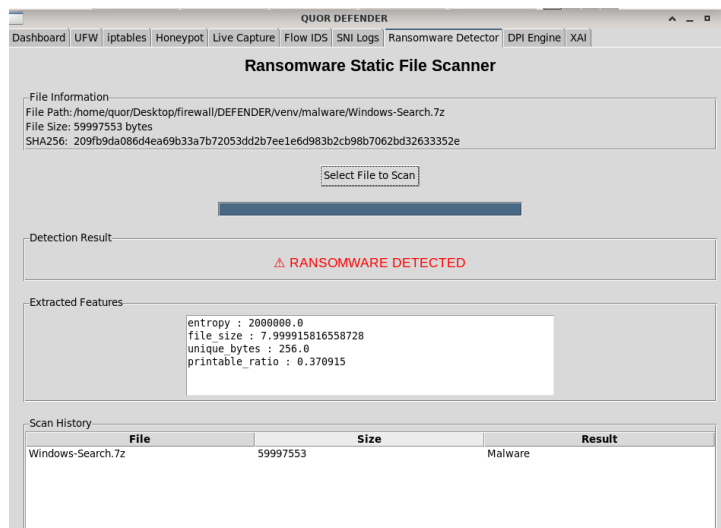


Figure 4.4: Ransomware Detection

4.7 Explainable AI (XAI) for Detection Transparency

To improve the transparency of the machine learning predictions, the system also incorporates an explainable AI component. Instead of simply classifying traffic as normal or malicious, the model can provide insight into which network features contributed most to the decision.

Feature importance values from the Random Forest model are used to understand how different traffic characteristics influence the prediction results. For example, abnormal packet sizes, unusual connection patterns, or excessive connection attempts may contribute to a malicious classification.

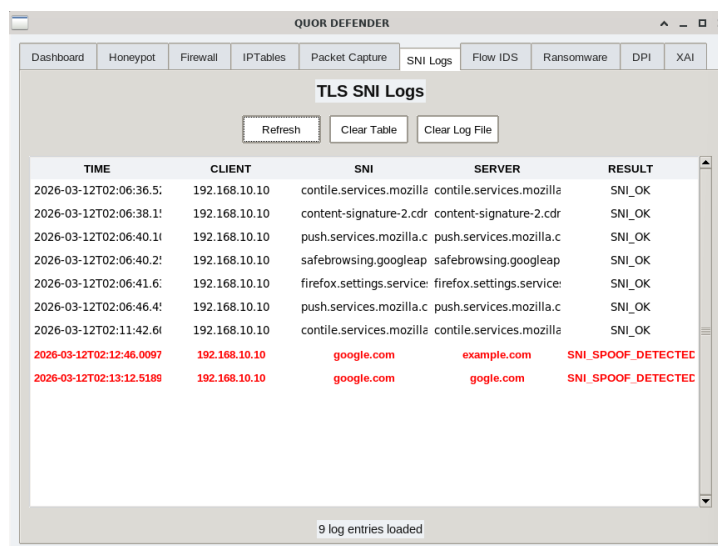
Providing explanations for the detection results helps administrators understand the behavior of the system and increases confidence in the machine learning based detection process. It also helps during system tuning and debugging when improving the detection accuracy.

4.8 TLS and SNI Inspection using Mitmproxy

Many modern cyber attacks use encrypted communication channels to hide malicious activities. Because encrypted HTTPS traffic cannot easily be inspected using traditional packet inspection methods, the system integrates mitmproxy to analyze TLS sessions.

Mitmproxy functions as an intercepting proxy that observes the TLS handshake process. Instead of fully decrypting the entire communication, the system focuses on analyzing metadata associated with the encrypted session. This includes information such as the Server Name Indication (SNI), certificate issuer, certificate validity period, and TLS version.

By examining this information, the firewall can detect suspicious encrypted connections, including possible spoofed certificates or abnormal domain communication patterns. This approach allows encrypted traffic to be analyzed without significantly affecting network performance.



TIME	CLIENT	SNI	SERVER	RESULT
2026-03-12T02:06:36.5:	192.168.10.10	contile.services.mozilla	contile.services.mozilla	SNI_OK
2026-03-12T02:06:38.1:	192.168.10.10	content-signature-2.cdr	content-signature-2.cdr	SNI_OK
2026-03-12T02:06:40.1:	192.168.10.10	push.services.mozilla.c	push.services.mozilla.c	SNI_OK
2026-03-12T02:06:40.2:	192.168.10.10	safebrowsing.googleap	safebrowsing.googleap	SNI_OK
2026-03-12T02:06:41.6:	192.168.10.10	firefox.settings.service	firefox.settings.service	SNI_OK
2026-03-12T02:06:46.4:	192.168.10.10	push.services.mozilla.c	push.services.mozilla.c	SNI_OK
2026-03-12T02:11:42.6:	192.168.10.10	contile.services.mozilla	contile.services.mozilla	SNI_OK
2026-03-12T02:12:46.0097	192.168.10.10	google.com	example.com	SNI_SPOOF_DETECTED
2026-03-12T02:13:12.5189	192.168.10.10	google.com	gogle.com	SNI_SPOOF_DETECTED

9 log entries loaded

Figure 4.5: SNI Spoof Detection

4.9 Honeypot Implementation

The system also includes a simple honeypot mechanism designed to observe attacker behavior. When repeated suspicious activity is detected from a particular source, the firewall can redirect that traffic to a honeypot service running on a trap port.

The honeypot simulates a vulnerable service in order to attract attackers and record their actions. Information such as the attacker's IP address, attempted commands, connection timestamps, and attack behavior are captured during this interaction.

This data can be useful for understanding attack patterns and improving future detection capabilities. The honeypot mechanism also prevents attackers from interacting with the actual system resources while still allowing the system to collect intelligence about their activity.

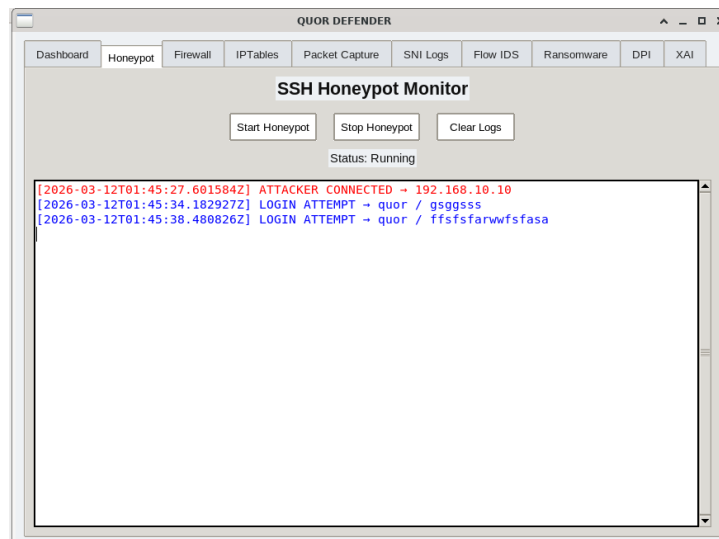


Figure 4.6: Honeypot Implementation

4.10 Logging and Monitoring

All important events generated during packet processing are recorded in the system logging module. Logging plays a critical role in monitoring the behavior of the firewall and analyzing security incidents.

The logging system records events such as blocked packets, suspicious TLS connections, machine learning detection results, and honeypot interactions. These logs can later be analyzed to identify patterns of attack or abnormal network behavior.

Maintaining detailed logs also helps administrators perform forensic analysis in the event of a security incident and allows the firewall detection mechanisms to be improved over time.

4.11 Implementation Workflow

The overall implementation follows a sequential packet processing workflow. First, network packets are captured from the network interface using Scapy. The captured packets are then checked against firewall rules to determine whether they should be allowed or blocked. Packets that pass this stage are further inspected using deep packet inspection techniques.

Next, the extracted packet features are analyzed by the machine learning model to identify abnormal traffic patterns. TLS connections are also inspected using mitmproxy to detect suspicious encrypted communication. If malicious behavior is detected, the system may block the traffic or redirect it to a honeypot trap. Throughout the entire process, important events are recorded in the logging system. This layered inspection approach ensures that multiple detection techniques work together to identify and mitigate potential cyber threats.

Chapter 5

Result and Evaluation

5.1 Results

The implemented QUOR firewall system was tested in a controlled environment to observe its behavior under both normal and suspicious network traffic conditions. The objective of the testing phase was to verify whether the different modules of the system such as packet capture, firewall rule enforcement, deep packet inspection, ransomware detection, TLS inspection, and honeypot monitoring were functioning correctly when integrated together.

During testing, network packets were successfully captured using the Scapy packet monitoring module. The system was able to extract important traffic features including source IP address, destination IP address, protocol type, packet length, and port numbers. These features were used for rule-based filtering as well as machine learning based analysis.

The firewall rule engine effectively blocked traffic that matched restricted ports or suspicious IP addresses. Packets that passed this stage were forwarded to the Deep Packet Inspection module, where payload analysis was performed to identify abnormal communication patterns.

The machine learning based detection module, implemented using a Random Forest classifier, analyzed the extracted network features and classified traffic as either normal or potentially malicious. The system was able to detect abnormal traffic patterns that resembled ransomware communication behavior during simulated testing.

Encrypted HTTPS traffic was analyzed using the mitmproxy module, which allowed inspection of TLS handshake metadata such as Server Name Indication (SNI) and certificate details. This helped the system monitor encrypted traffic without directly decrypting the payload.

The honeypot component was also tested by redirecting suspicious traffic to a simulated trap service. The honeypot successfully recorded attacker interaction details such as connection attempts and source IP addresses. All detected events were logged in the

system for monitoring and analysis.

Overall, the testing results indicate that the system is capable of monitoring network traffic and identifying suspicious behavior using a layered detection approach.

5.2 Evaluation

The performance of the QUOR firewall system was evaluated by measuring the accuracy of the machine learning detection module and the time taken for packet analysis during different test runs. Multiple experiments were conducted using network traffic samples containing both normal and abnormal traffic patterns.

The Random Forest classifier showed good performance in identifying suspicious traffic behavior based on the extracted network features. Detection results were analyzed using metrics such as accuracy and processing time. The evaluation indicates that the system can detect abnormal traffic patterns with reasonable accuracy while maintaining acceptable processing speed.

In addition to machine learning evaluation, the effectiveness of rule-based filtering and deep packet inspection was also observed. The firewall rules were able to block unauthorized connections immediately, reducing the number of packets that required deeper analysis. TLS inspection using mitmproxy also provided useful information about encrypted sessions, which improved overall traffic visibility.

The combined use of rule-based filtering, deep packet inspection, machine learning analysis, and honeypot monitoring allowed the system to identify suspicious activities more effectively than a single-layer detection approach.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presented the design and implementation of a virtualized Next Generation Firewall system called QUOR. The primary objective of the project was to improve network security by combining traditional firewall mechanisms with intelligent traffic analysis techniques. The system was developed using a modular approach so that different security components could work together to monitor and analyze network activity.

The implemented firewall system integrates several features such as packet capture, rule-based filtering, deep packet inspection, machine learning based ransomware detection, TLS inspection using mitmproxy, explainable AI analysis, and a honeypot monitoring mechanism. Each of these components contributes to identifying suspicious traffic and protecting the internal network from potential threats.

During testing, the system was able to capture network traffic, apply firewall rules, inspect packet contents, and detect abnormal communication patterns. The machine learning model helped identify potentially malicious traffic, while the honeypot component allowed observation of suspicious connection attempts. TLS inspection also provided visibility into encrypted connections without directly decrypting the traffic.

Overall, the results demonstrate that combining multiple security techniques within a single firewall framework can improve network monitoring and threat detection. The project provides a practical prototype that illustrates how intelligent firewall systems can be implemented using open source tools and machine learning techniques.

6.2 Future Work

Although the current implementation demonstrates the basic functionality of the QUOR firewall system, several improvements can be made to enhance its capabilities in the future.

One possible improvement is expanding the machine learning detection module by training models on larger and more diverse network traffic datasets. This could improve the accuracy of anomaly detection and allow the system to recognize a wider range of attack patterns.

Another potential enhancement involves optimizing the deep packet inspection module to improve performance when handling high volumes of network traffic. Efficient packet processing techniques could help the system operate more effectively in larger network environments.

Future work may also include the development of a more advanced monitoring interface that provides graphical visualization of traffic patterns, alerts, and detection statistics. This would make it easier for administrators to analyze system activity and manage firewall policies.

Additionally, integrating automated threat intelligence feeds and dynamic rule updates could further strengthen the system's ability to respond to emerging cyber threats. These improvements would help extend the current prototype into a more comprehensive and scalable network security solution.

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